Category
Equity Valuation: Concepts and Basic Tools

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ENTERPRISE VALUE

Learning Outcomes

- describe enterprise value multiples and their use in estimating equity value
- explain advantages and disadvantages of each category of valuation model

An alternative to estimating the value of equity is to estimate the value of the enterprise. Enterprise value is most frequently determined as market capitalization plus market value of preferred stock plus market value of debt minus cash and investments (cash equivalents and short-term investments). Enterprise value is often viewed as the cost of a takeover: In the event of a buyout, the acquiring company assumes the acquired company's debt but also receives its cash. Enterprise value is most useful when comparing companies with significant differences in capital structure.

Enterprise value (EV) multiples are widely used in Europe, with EV/EBITDA arguably the most common. EBITDA is a proxy for operating cash flow because it excludes depreciation and amortization. EBITDA may include other non-cash expenses, however, and non-cash revenues. EBITDA can be viewed as a source of funds to pay interest, dividends, and taxes. Because EBITDA is calculated prior to payment to any of the company's financial stakeholders, using it to estimate enterprise value is logically appropriate.

Using enterprise value instead of market capitalization to determine a multiple can be useful to analysts. Even where the P/E is problematic because of negative earnings, the EV/EBITDA multiple can generally be computed because EBITDA is usually positive. An alternative to using EBITDA in EV multiples is to use operating income.

In practice, analysts may have difficulty accurately assessing enterprise value if they do not have access to market quotations for the company's debt. When current market quotations are not available, bond values may be estimated from current quotations for bonds with similar maturity, sector, and credit characteristics. Substituting the book value of debt for the market value of debt provides only a rough estimate of the debt's market value. This is because market interest rates change and investors' perception of the issuer's credit risk may have changed since the debt was issued.

EXAMPLE 16

Estimating the Market Value of Debt and Enterprise Value

1. Cameco Corporation is one of the world's largest uranium producers; it accounts for 16 percent of world production from its mines in Canada and the United States. Cameco estimates it has about 458 million kilograms of proven and probable reserves and holds premier land positions in the world's most promising areas for new uranium discoveries in Canada and Australia. Cameco is also a leading provider of processing services required to produce fuel for nuclear power plants. It generates 1,000 megawatts of electricity through a partnership in North America's largest nuclear generating station located in Ontario, Canada.

   For simplicity of exposition in this example, we will present share counts in thousands and all dollar amounts in thousands of Canadian dollars. In 2017, Cameco had 395,793 shares outstanding. Its 2017 year-end share price was $14.11. Therefore, Cameco's 2017 year-end market capitalization was $5,584,640.

Year	Payment
2018	$69,000
2019 and 2020	610,000
2021 and 2022	482,000
Thereafter	744,000
Total	$1,905,000

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Cameco's longest maturity debt matures in 2042. We will assume that the amounts paid in 2019 and 2020, and in 2021 and 2022, will be paid equally during the two years. The "thereafter" period includes two debenture tranches, the first one maturing in 2024 for a total value of $620,000 and the second tranche maturing in 2042 for the remaining $124,000. A yield curve for zero-coupon Canadian government securities was available from the Bank of Canada. The yield-curve data and assumed risk premiums in Exhibit 9 were used to estimate the market value of Cameco's long-term debt:

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Exhibit 9:

Estimated Market Value

Year	Yield on Zero-Coupon Government Security (%)	Assumed Risk Premium (%)	Discount Rate (%)	Book Value	Market Value
2018	0.89	0.50	1.39	$69,000	$68,054
2019	1.11	1.00	2.11	$305,000	$292,525
2020	1.39	1.50	2.89	$305,000	$280,014
2021	1.65	2.00	3.65	$241,000	$208,804
2022	1.88	2.50	4.38	$241,000	$194,505
2023	2.10	3.00	5.10	$0	$0
2024	2.30	3.50	5.80	$620,000	$417,823
2025	2.50	4.00	6.50	$0	$0
…	…	…	…	$…	$...
2042	2.92	5.00	7.92	$124,000	$18,445
				$1,905,000	$1,480,170

Note from Exhibit 9 that the book value of long-term debt is $1,905,000 and its estimated market value is $1,480,170. The book value of total debt and liabilities of $2,919,100 minus the book value of long-term debt of $1,905,000 is $1,014,100. If we assume that the market value of that remaining debt is equal to its book value of $1,014,100 an estimate of the market value of total debt and liabilities is that amount plus the estimated market value of long-term debt of $1,480,170 or $2,494,270.

At the end of 2017, Cameco had cash and equivalents of $591,600. Enterprise value can be estimated as the $5,584,640 market value of stock plus the $2,494,270 market value of debt minus the $591,600 cash and equivalents, or $7,487,310. Cameco's 2017 EBITDA was $606,000; an estimate of EV/EBITDA is, therefore, $7,487,310 divided by $606,000, or 12.4.

EXAMPLE 17

information in Exhibit 10, which two mining companies seem to be the *most* undervalued?

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Exhibit 10: Data for Twelve Major Mining Companies

Company	EV (US$ millions)	Operating Income (OI) (US$ millions)	EV/OI
BHP Billiton	119,712.3	11,753	10.19
Rio Tinto	93,856.1	6,471	14.5
Vale	82,051.2	6,366	12.89
Glencore	80,772.0	–549	–147.13
Southern Copper	37,817.0	1,564	24.18
Freeport-McMoRan	33,452.0	–2,766	–12.09
Anglo American	32,870.3	2,562	12.83
Norilsk Nickel	22,483.0	3,377	6.66
Coal India	21,652.1	1,382	15.67
Barrick Gold	21,549.8	2,424	8.89
Newmont Mining	20,683.0	–65	–318.20
Goldcorp	12,986.7	369	35.19

Source: www.miningfeeds.com, Morningstar.

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Hide Solution

Solution:

Norilsk Nickel and Barrick Gold have the lowest EV/OI and thus appear to be the *most* undervalued or favorably priced on the basis of the EV/OI. Note the negative ratio for Glencore, Freeport-McMoRan, and Newmont Mining. Negative ratios are difficult to interpret, so other means are used to evaluate companies with negative ratios.

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